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Some categorical properties in terms of operads

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A type of algebras whose relations are multilinear can be seen as a category of algebras over an operad ([1]). From this point of view, we can understand operads as algebraic structures in which we can encode the information of the operations defined in our algebras. We are interested in understanding the link between the properties of an operad and the categorical properties of the category of algebras over it. It is known that, as soon as we have no constants different from 0 in our algebras, we are in a semi-abelian setting. We show an explicit proof of this fact.

Once that is done, we go forward and study some categorical properties in this context. In particular, we express in terms of operads the properties Normality of Higgins (NH), Smith is Huq (SH) and Algebraic Coherence (AC). We find that in the particular case of algebras over a reduced operad it is equivalent satisfying AC and both NH and SH. We also study Locally Algebraically Cartesian Closedness (LACC), finding for this context a strong condition in our operads.

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References

[1] J. Loday and B. Vallette, *Algebraic Operads*, volume 346 of *Grundlehren der Mathematischen Wissenschaften [Fundamental Principles of Mathematical Sciences]*. Springer, Heidelberg, 2012.